NACS 645 – Moral technologies

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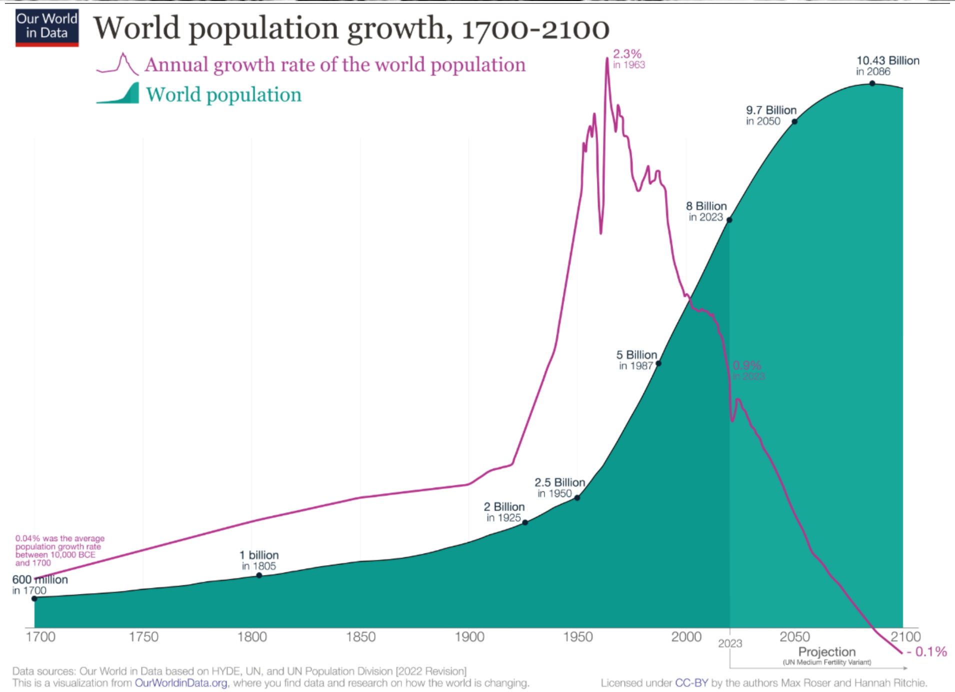
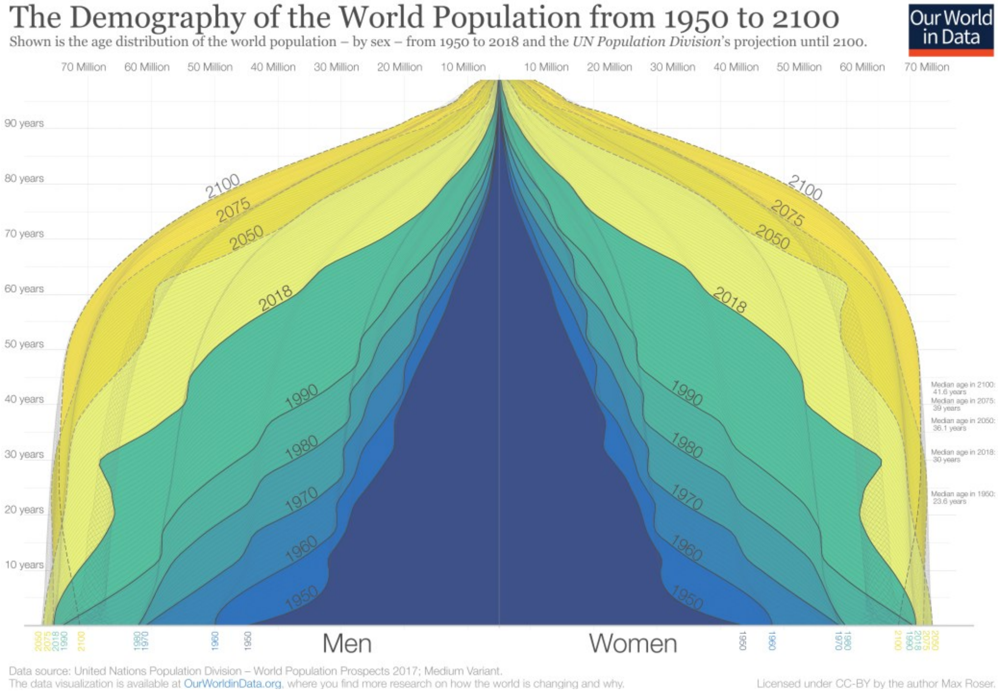
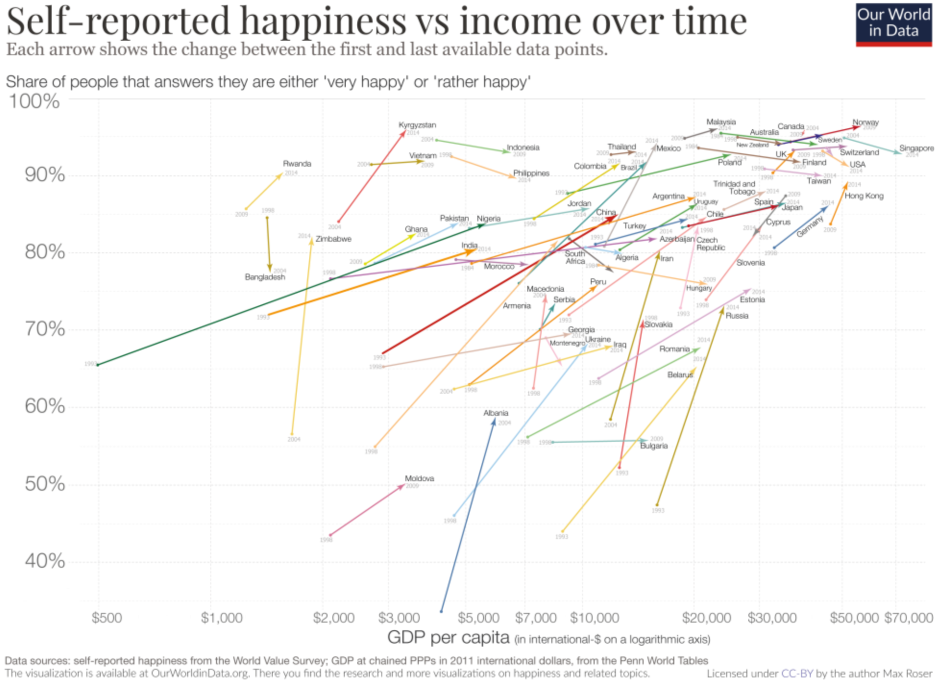
Population explosion



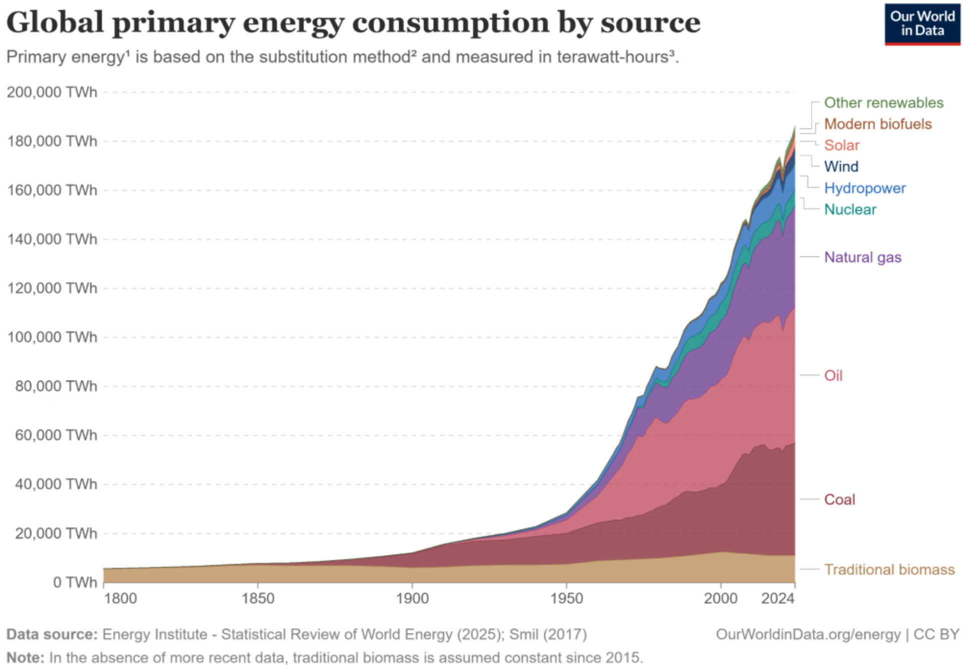
Higher life expectancy



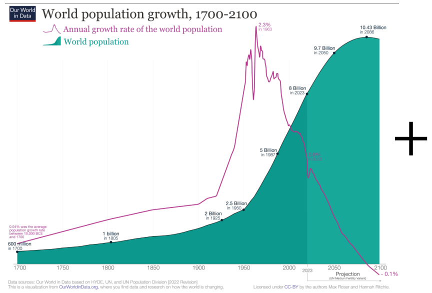
Progressing income and happiness

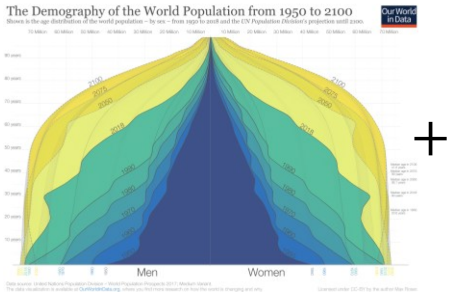


Stronger energy consumption

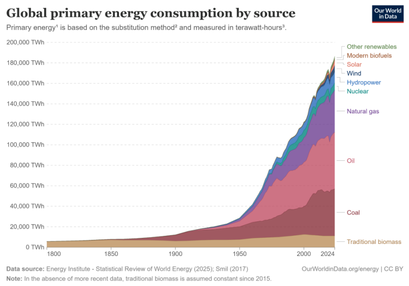


Common Good maximization problem

$Good = \omega_1 \cdot$ 

$+\omega_2 \cdot$ 

$+\omega_3 \cdot$ 

$+\omega_4 \cdot$ 

$+\varepsilon$

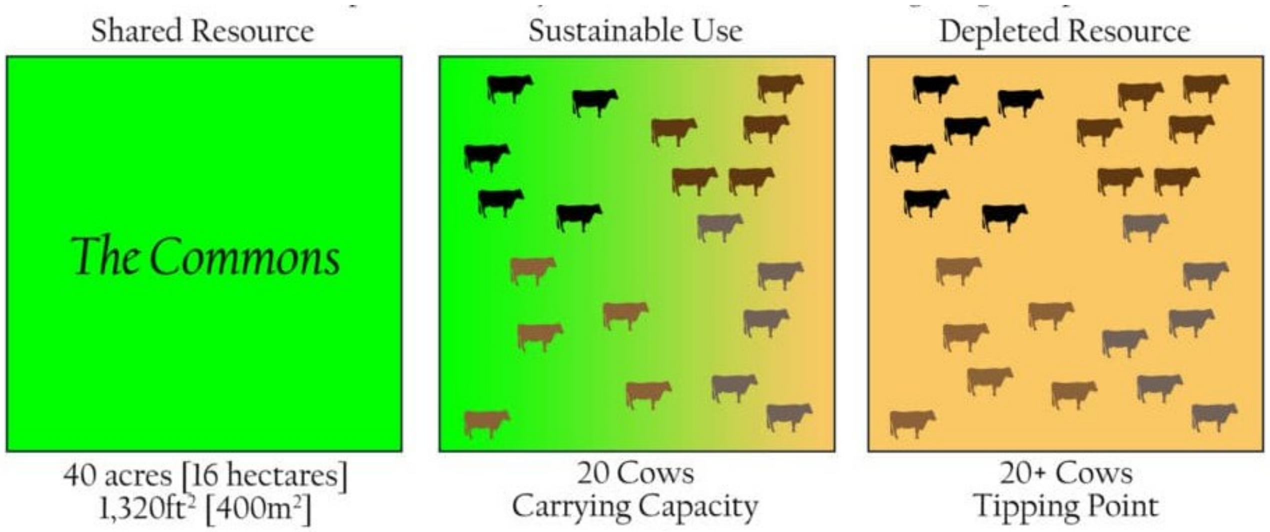
Hardin:

- In a finite world, *maximizing population* requires driving ***work calories per person toward zero***: no art, no gastronomy, no leisure
- What we gain in numbers, we lose in quality of life
- ***Technical solution*** (infinite energy supply) **only shifts** *energy acquisition problem* to *energy dissipation problem*
- **We can't jointly maximize two variables** (e.g., population, welfare): *mathematically and materially incompatible*
- Hence, the task is not to **maximize**, but to **optimize**, within constraints, the *Good* we choose

- **Technologies lift constraints** and rise ceilings, **but do not remove the maximization problem**
- **Optimizing the *Good*** in a finite world therefore demands a ***non-technical solution***
- It also creates a resource allocation problem

Tragedy of the commons

Pasture open to all (i.e., commons):
- Each herdsman can **increase private utility** by adding more cattle
- Finite resource, but unrestricted access



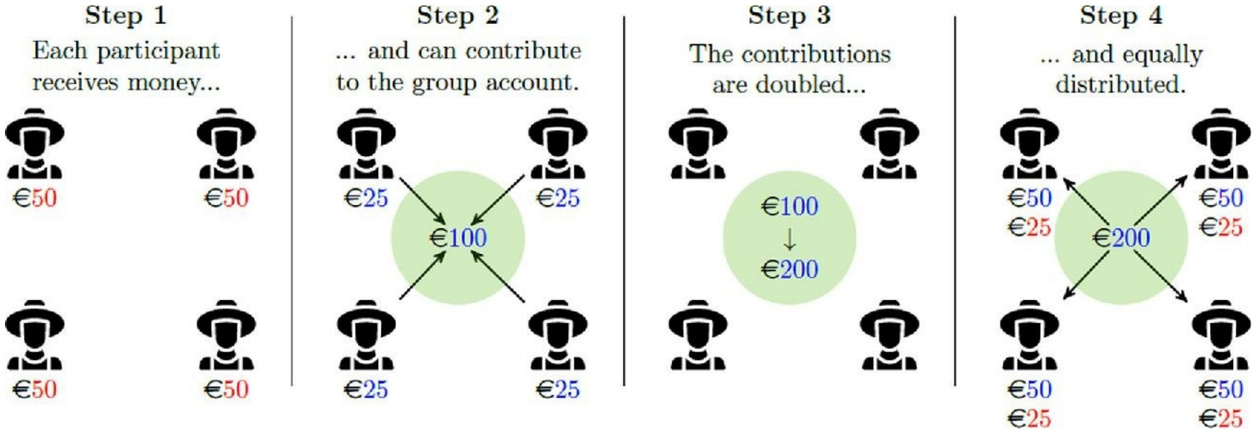
Adapted from: Ric Stephens, University of Oregon College of Design, School of Planning, Public Policy, and Management.

Utility Calculation:
- **Positive component: +1** (private gain from the extra cow)
- **Negative component: −ε** (fractional share of the collective loss from overgrazing)

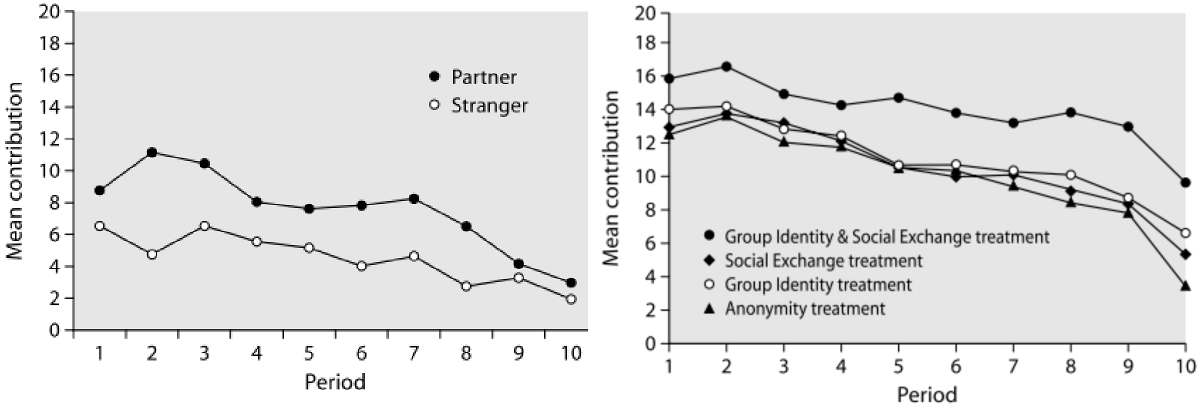
The Tragedy:
- Since $\varepsilon \ll 1$, rational choice for each is to add another cow…
  until pasture is degraded beyond recovery
- **Individually rational actions aggregates into collective ruin**

Implications:
- The commons is a **non-zero-sum game of mixed cooperative and competitive motives**
- In a finite system, **individual liberty and collective welfare cannot both be maximized**
- Technologies may lift constraints but cannot alter the incentive structure that rewards self-interest
- Avoiding tragedy requires **collective rules, limits, and moral restraint**

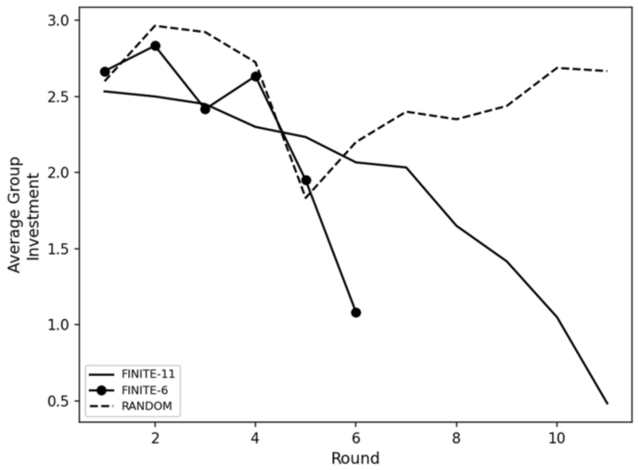
Public goods games



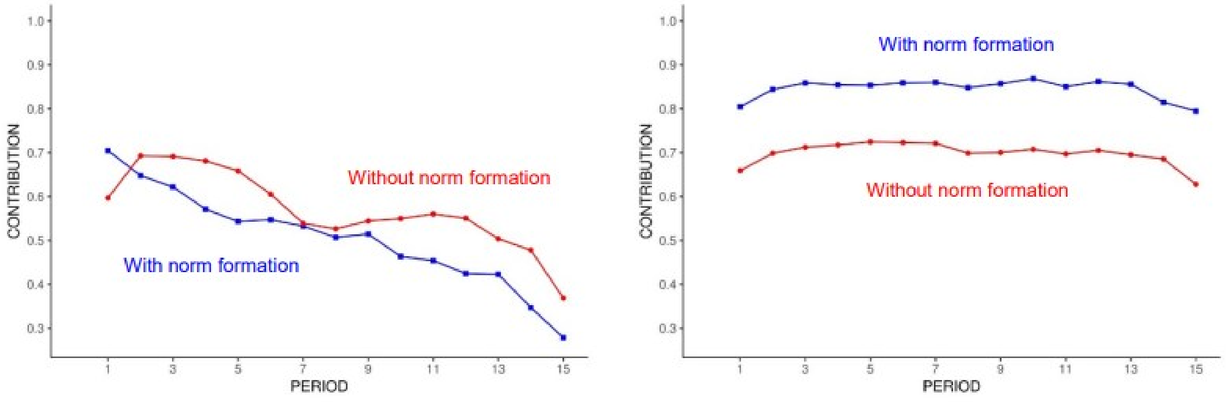
Adapted from Rommel et al., 2022. *Q Open*.



Adapted from Gächter & Herrmann, 2006. Cooperation in primates and humans: mechanisms and evolution



Adapted from Anderson et al., 2024. *Journal of the Economic Science Association*



(a) Public goods game without peer punishment

(b) Public goods game with peer punishment

Adapted from Fehr & Schurtenberger, 2018. *Nature Human Behavior*.

Greene & Young, 2020. The Cognitive Neuroscience of Moral Judgment and Decision-Making
Greene, 2014. *Penguin Press.*
Greene, 2015. *Cognition.*

The moral cognition

Morality is a collection of biological and cultural solutions that promote and sustain cooperation, not special organ or processes.

Moral cognition integrates valuation, reasoning, control, emotions to regulate cooperative behavior.

Moral emotions operate as *psychological carrots and sticks*.

e.g., Guilt → self-directed punishment; *Gratitude* → reward toward others

- Moral emotions (empathy, guilt, righteous anger, compassion) act as internalized regulators: They encode explicit moral motives ("I should," "That's wrong")
- Non-moral social emotions (gossip, embarrassment, vengefulness, in-group favoritism) act as indirect or local regulators: They manage reputation, kinship, coalitions, but are not moral obligations

Neural architecture:

- No dedicated "moral" module; relies on brain systems applied to cooperative problems :
  - Value and motivation – vmPFC, striatum
  - Emotion and affective learning – amygdala, insula
  - Mental-state representation – TPJ, mPFC
  - Cognitive control – dlPFC, ACC
  - Simulation and imagination – hippocampus, DMN

Implications:

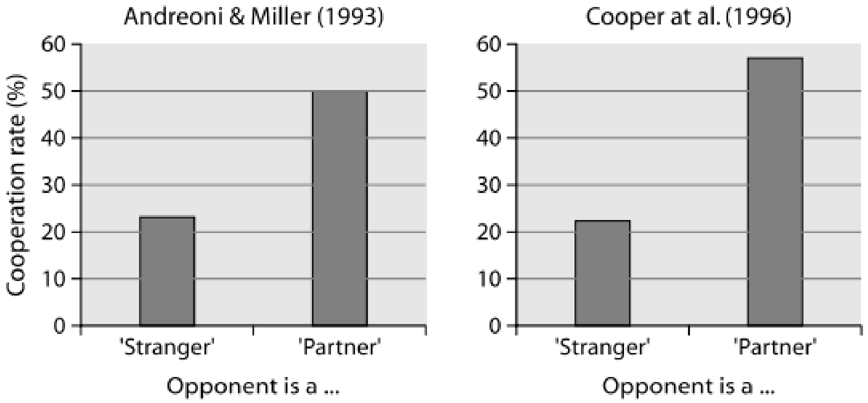
Within this framework:

- The *moral* character of a behavior/process comes from its *cooperative* function
- The "moral brain" is the brain's distributed network tackling cooperative problems through its ordinary systems for value, emotion, control, and social cognition

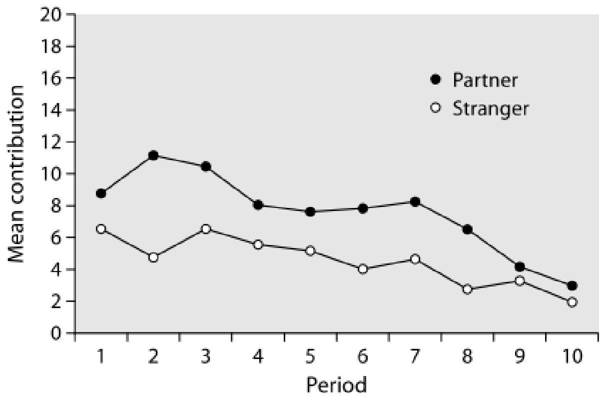
Gächter & Herrmann, 2006. *Cooperation in primates and humans: mechanisms and evolution.*
Fehr & Schurtenberger, 2018. *Nature Human Behavior.*

Maintaining cooperation

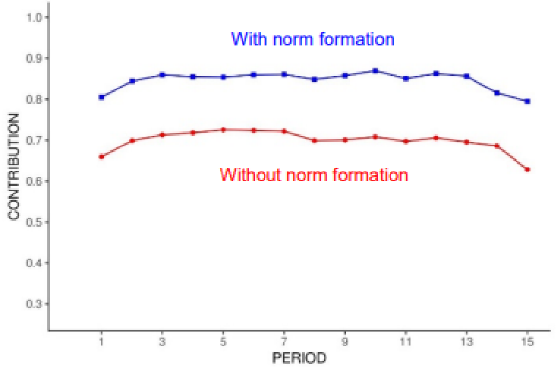
Despite overgrazing, tax evasion, or failed agreements on environmental protection, humans achieve high levels of cooperation – even among strangers



Prisoner's Dilemma with constant (Partner) and randomly-changing (Stranger) opponents



Public good game with constant (Partner) and randomly-changing (Strangers) groups



(b) Public goods game with peer punishment

1. Evolutionary Pathways for Cooperation
(selection mechanisms for cooperation; cf. Rand & Nowak, 2013)

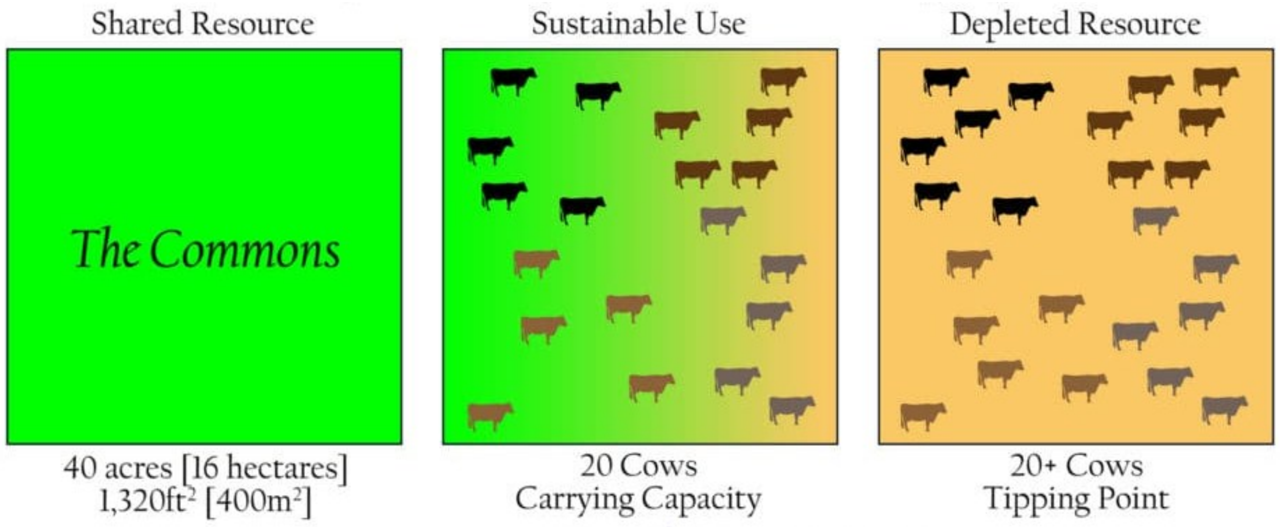
- **Kin selection**
- **Direct reciprocity**
- **Indirect reciprocity**
- **Spatial selection**
- **Group selection**

2. Proximate and Institutional Reinforcers
(mechanisms for cooperation within societies)

- **Altruistic punishment** – Costly sanctioning of free riders
- **Moral emotions** – Anger, guilt, shame, and empathy as immediate motivators for punishment and compliance
- **Social norms** – *Known standards of behavior based on shared beliefs about how individuals ought to behave in a specific context*
- **Rules and institutions** – Formal codifications of norms with enforcement mechanisms

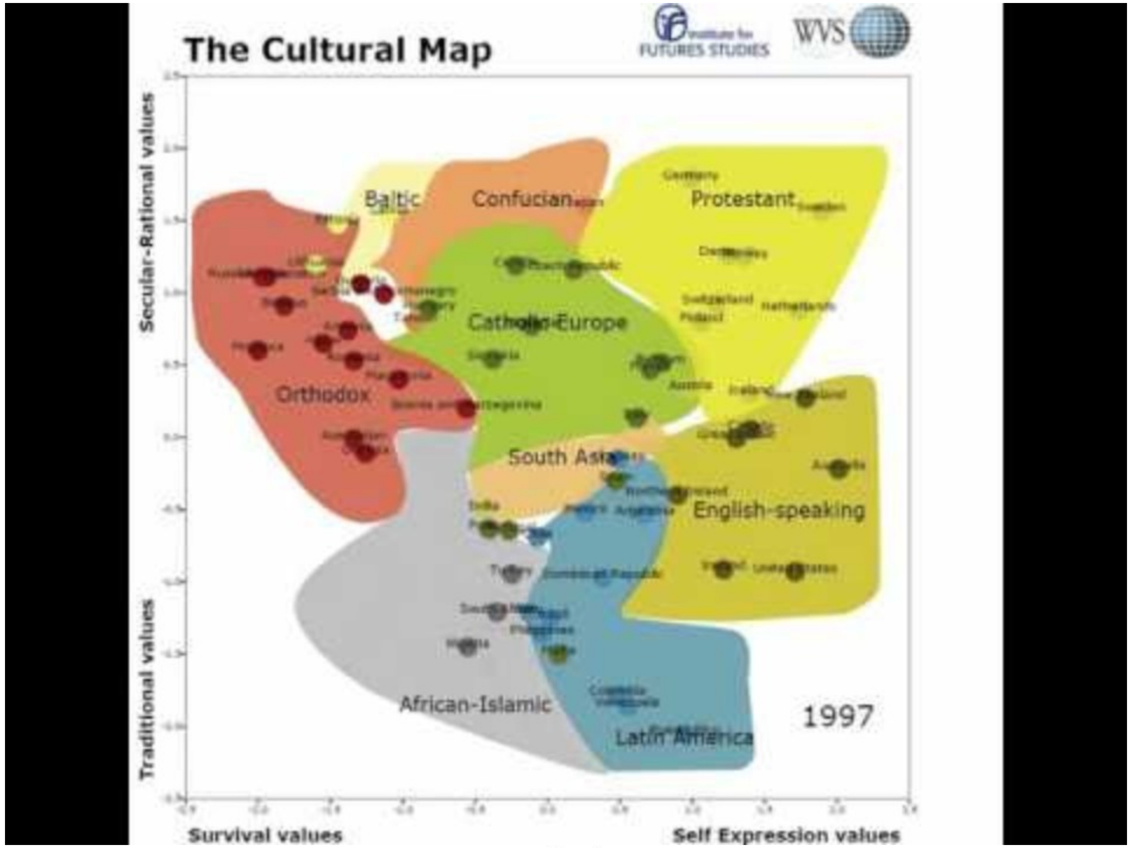
Cooperation at the population-level

Joshua Greene, Moral Tribes (2013). *Penguin Press*



- Hardin: We need *mutual coercion mutually agreed upon*
- Rand, Nowak: Evolution gives produces genuine cooperators – intuitive reciprocation and selective mechanisms sustain coop.
- Greene; Curry: *Human morality evolved* to align individual self-interest with collective welfare
  - Shared norms, emotions, reputations make selfish behavior costly. Guilt, shame, pride, and gratitude act as internal "moral technologies"
  - Emotions regulate local cooperation; reason, norms and institutions extend cooperation to the group level

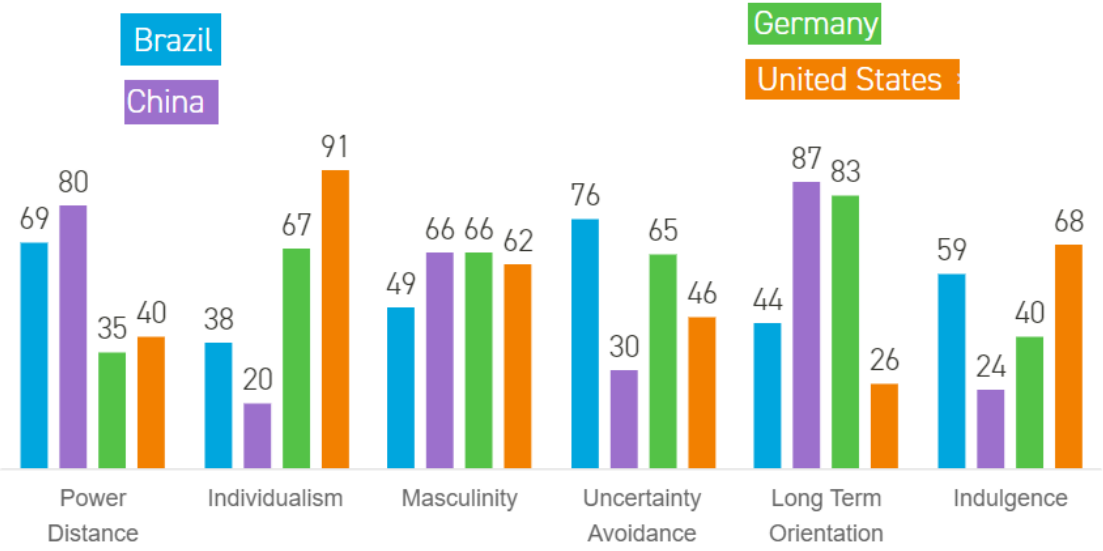
Cultural differences



Inglehart-Welzel culture map

1. *Traditional vs secular–rational values*:
   traditional values emphasize religion, parent–child ties, deference to authority; secular-rational values emphasize more acceptance of divorce, abortion, euthanasia, etc.

2. *Survival vs self-expression values:*
   survival values emphasize economic/physical security, ethnocentrism, low trust/tolerance; self-expression values emphasize subjectively defined well-being, tolerance of diversity, participatory decision-making, etc.



Hofstede's cultural dimensions theory

- **Power distance**: describes the extent to which less powerful members of a society accept and expect unequal distribution of power.
- **Individualism vs collectivism**: measures whether people in a culture prioritise personal goals over group goals, or vice versa.
- **Masculinity vs femininity**: Masculinity refers to cultures that value competitiveness, achievement, and material success, while femininity values cooperation, care, and quality of life.
- **Uncertainty avoidance**: measures how comfortable a culture is with ambiguity, change, and the unknown.
- **Long-term avoidance**: reflects whether a culture prioritises future rewards over immediate results.
- **Indulgence vs restraint**: looks at how freely societies allow people to gratify their desires and enjoy life.

Kitayama & Uskul, 2011. *Annual Review of Psychology.*
Chiao et al., 2013. Psychological Inquiry.

Culture, Mind, and the Brain

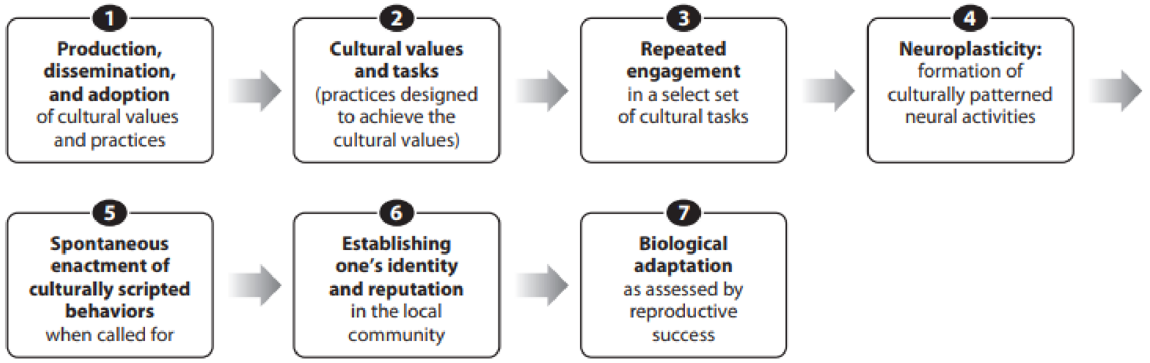


Figure 1

A neuro-culture interaction model. Values and practices of culture are produced, disseminated, and adopted as a function of a variety of collective-level factors. Individuals choose some select set of available cultural practices as their own cultural tasks. They then actively engage in them so as to realize their culture's primary values such as independence and interdependence in their own idiosyncratic ways. Repeated engagement in the cultural tasks results in culturally patterned brain activities, which in turn enable the individuals to spontaneously and seamlessly enact the culturally scripted behaviors when such behaviors are called for by situational norms. The ability of the individuals to perform the culturally scripted behaviors when normatively required to do so enhances their own identity and reputation as a decent member of the cultural tradition and, eventually, their ability to achieve biological adaptation as assessed by reproductive fitness.

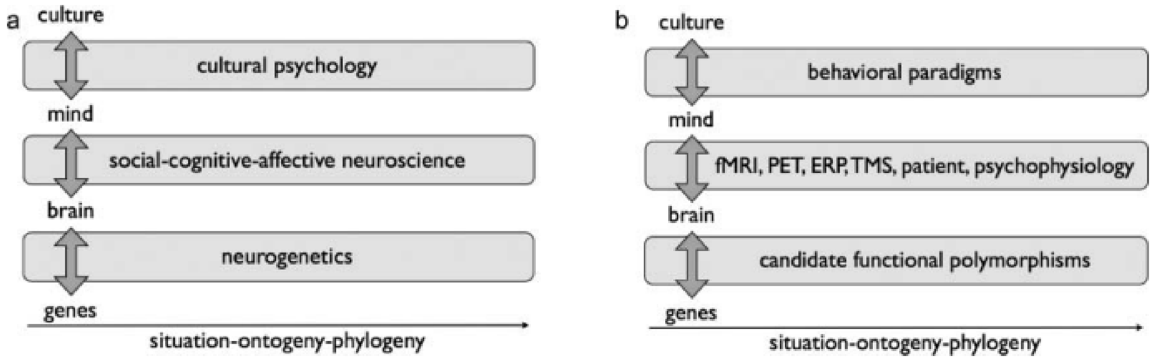


Figure 1. Framework of cultural neuroscience (Chiao, 2009, 2011; Chiao & Ambady, 2007). *Note.* (a–b) Research in cultural neuroscience integrates theory and methods from cultural psychology, social-cognitive-affective neuroscience and neurogenetics across multiple time scales, specifically situation, ontogeny and phylogeny. *Source.* Adapted from Chiao and Ambady (2007) and Chiao (2009).

If we assume the same set of mechanisms for the evolution of cooperation,
should we observe cooperation to the (relatively) same extent in distinct societies?